

Sujal Kumar Kanu

Computer Science Engineering Student | Full-Stack · Blockchain · Generative AI

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TECHNICAL SKILLS

- Languages**
C, C++, Java, JavaScript (ES6+), Python, HTML5, CSS3, SQL, Solidity
- Frameworks**
React.js, Node.js, Express.js, FastAPI, TailwindCSS, Ethers.js, Recharts
- Tools & Databases**
PostgreSQL, Redis, Hardhat, Docker, Git & GitHub, Linux (Fedora/Ubuntu), MATLAB, VS Code
- Domains**
Data Structures & Algorithms, Full-Stack Development, Smart Contracts & Blockchain, 3D Web Graphics, IoT
- Soft Skills**
Problem Solving, Team Collaboration, Technical Presentations, Rapid Prototyping

CERTIFICATIONS

- Introduction to Cyber Security**
Infosys Springboard · Jun 2025
- Statistics Onramp & Curve Fitting Onramp**
MathWorks, 100% completion · Oct 2025
- IoT & Robotics, 7-Day Intensive**
NepaTronix x National Infotech · 2023

INTERESTS

- Generative AI: LLMs, Prompt Engineering, AI Agents
- Ethical hacking & network security
- Machine learning & statistical modelling (Python, MATLAB)

LANGUAGES

- Bhojpuri** · Native
- Hindi** · Fluent
- Nepali** · Fluent
- English** · Professional

PROFILE

Third-year Computer Science student building end-to-end products across blockchain, full-stack web and applied AI. Hands-on with Solidity, React, FastAPI and Docker, with national-level hackathon experience and a strong DSA and systems foundation.

EDUCATION

- BMS Institute of Technology & Management (VTU)** 2023 – Present
B.E. in Computer Science & Engineering (3rd Year), Bengaluru
Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Organization, Discrete Mathematics
- National Infotech Secondary School, Birgunj** 2024
Grade XII (NEB, Nepal)

PROJECTS

- AgriChain: Blockchain Supply Chain & Fraud Detection** Polygon Amoy · Docker
React.js, Node.js/Express, Solidity, Polygon Amoy Testnet, Ethers.js, Docker Compose
- Engineered a transparent 4-tier lifecycle tracker (Farmer, Distributor, Retailer, Consumer) with Solidity smart contracts for immutable records.
 - Built a 3-layer fraud detection engine flagging duplicate lots, sequence gaps and price spikes (>500%) with dynamic trust scores (0–100).
 - Integrated batch-level QR code provenance verification, AfterShip shipment monitoring and strict blockchain execution mode.

- Research Collab Hub: AI Paper & Resource Platform** FastAPI · PostgreSQL
FastAPI, React 18, PostgreSQL 16, Redis, Anthropic Claude API, Docker
- Built a modular collaboration platform with automated paper versioning triggers, BibTeX citation export and conference deadline tracking.
 - Integrated Claude AI inference pipelines for abstract decomposition, venue suitability analysis and co-author matching.

- CardioSim 3D: Heart Anatomy & Pathology Simulation** Vanilla JS · WebGL
Vanilla JavaScript (ES6+), HTML5, CSS3 Glassmorphism, Sketchfab Viewer API
- Developed an interactive 3D heart simulation with orbit/zoom controls and callouts across 24+ annotated anatomical structures.
 - Synchronized cardiac pacing (30–200 BPM) with UI pulsing, condition simulations (arrhythmia, blocked artery) and a quiz mode.

- Weather Monitoring System (IoT)** 2023
- Built a real-time device with temperature, humidity and pressure sensors; presented to faculty and mentors.

- Statistical Modelling & Curve Fitting (MATLAB)** Oct – Nov 2025
- Applied probability distributions, regression and curve fitting to real-world datasets with residual analysis.

HACKATHONS & COMPETITIONS

- CODE RED 3.0**, National 24-Hour Hackathon, E-Cell BMSIT&M 2026
- Delivered a working solution in 24 hours, honing rapid prototyping, stress-testing and presentation skills.
- ArtPark CodeForge, Rhapsody 4.0**, IISc Bangalore Mar 2026
- Competed among 2,796+ teams (prize pool Rs. 50,000) in a national-level hackathon turning ideas into working prototypes.